

Unit 1: Intro to Statistics and R

Study Guide

After you're done: After the in-class discussions, how do you rate your grasp of these materials:

- [] 1: confused
 - [] 2: some things are clear, but not all
 - [] 3: ok, but need to study more
 - [] 4: piece of cake
-

Instructions

- Either before, during, or after engaging with the readings and recorded lectures, answer the concept checks below.
 - Date for in class discussion: Wed Sep 9.
 - The key points section will be discussed in class to clarify points of confusion. Bring it to class!
 - Note: If questions remain about these after class, please attend office hours to clarify them.
-

Assigned learning Materials

- Ch.1 + Interleaf 1 + Interleaf 6 from textbook.
- Lecture videos starting with “1.” (Moodle).
- Quiz questions at the end of lecture videos.
- Problem set 1

Other practice problems and learning materials:

- End of chapter questions (ch.1)
-

Learning Goals

By the end of this unit, you should be able to:

- Understand the major goals of statistics.
- Distinguish between a sample and a population.
- Distinguish between an estimate and a parameter.
- Identify why estimates from samples may deviate from the parameters of populations.
- Identify the properties of a good sample.
- Describe the differences between observational and experimental studies.
- Identify whether a variable is numeric or categorical (and subtypes).
- Distinguish between sampling bias and sampling error.
- Distinguish between accuracy and precision and how these terms relate to sampling bias and sampling error.
- Understand the distinction between the term “population” in a statistical and a biological context.
- Distinguish between correlation and causation.
- Understand why experimental studies reveal causation, while observational studies cannot. Recognize that confounding factors limit our ability to interpret observed patterns.

Key terms

You should be able to explain and apply these concepts in their relevant contexts. Try to define them below with your own words. Then look them up and check.

Population (statistical):

Sample:

Parameter:

Estimate:

Sampling error:

Sampling bias:

Independence:

Explanatory variables:

Response variables:

Quantitative variables:

Continuous variables:

Discrete variables:

Categorical variables:

Ordinal variables:

Nominal variables:

Key points

Try to answer these yourselves. Bring them to class to discuss and clarify point of confusion.

- 1) What are the some of the key goals of statistics?
- 2) Compare and contrast the different meanings of the word “population” (biological vs. statistical)
- 3) Why is sampling such a crucial concept in statistics?
- 4) Define and contrast these two concepts: parameters and estimates (statistics). Which of these is prone to bias and sampling error?
- 5) How can we reliably decrease sampling error?
- 6) Is a study involving lab work always experimental? Is a field study always observational?
- 7) Define and contrast probability distributions from frequency distributions. How does this relate to one of the key goals of statistics?
- 8) How can experiments introduce bias? How can bias be avoided in experiments?
- 9) Define and name the properties of a good sample.
- 10) Define and compare the types of variables in an experiment.

- 11) Discuss the relationship between statistical association, correlation, and causality. Come up with an example different from the textbook and lecture ones.
- 12) Does a causal relationship in a controlled experiment under specified settings always imply a causal relationship in nature?

References

- Brandvain, Y. (2022). Applied Biostats. Bookdown. https://bookdown.org/ybrandvain/Applied_Biostats_2022/
- Whitlock, M., & Schluter, D. (2020). The analysis of biological data. Macmillan. <https://whitlockschluter3e.zoology.ubc.ca/>